Linear Regression

Plan

1. Linear regression
   - A geometrical interpretation
   - A probabilistic intepretation
2. Regularization
   - ℓ_2 regularization (Ridge regression)
   - ℓ_1 regularization (Lasso regression)

Linear Regression

- Similar to classification, we are given:
  - $\mathbf{X} : n \times d$ the training data matrix
  - $\mathbf{y} : n \times 1$ the target vector, where now $y_i \in R$
- and we want to find a linear model $\mathbf{w}$ such that:

$$y_i = \mathbf{w}^T \mathbf{x}_i$$

or equivalently:

$$\mathbf{y} = \mathbf{X}\mathbf{w}$$

- However most of the times there will not be an exact solution
- thus as in classification we need a measure of error

Linear Regression error function

- Probably the most well known and often used error function in learning

$$(y - z)^2, \text{ where } z = \mathbf{w}^T\mathbf{x}$$

  the *squared error*

- thus the error over the training set is given by:

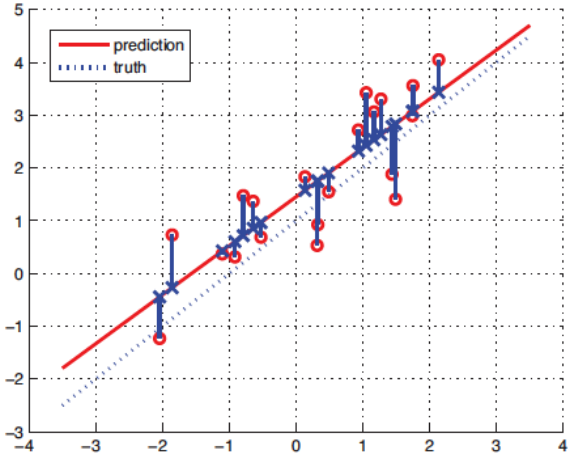
$$E(\mathbf{w}) = \sum (y_i - \mathbf{w}^T\mathbf{x}_i)^2 = ||\mathbf{y} - \mathbf{X}\mathbf{w}||^2 = (\mathbf{y} - \mathbf{X}\mathbf{w})^T(\mathbf{y} - \mathbf{X}\mathbf{w})$$

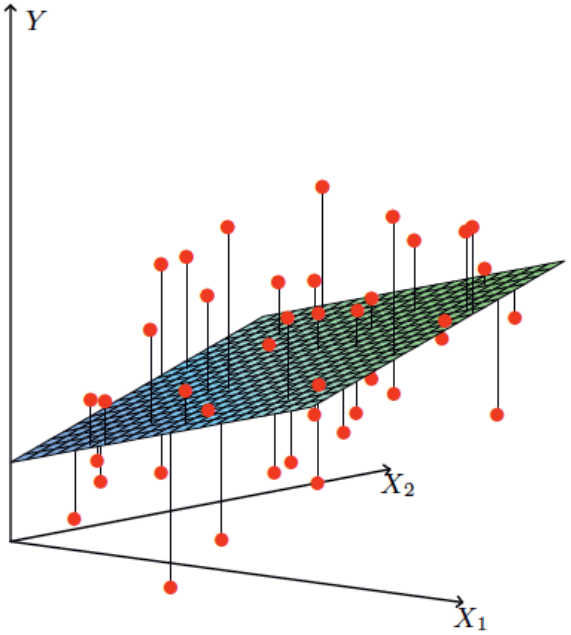
- and as before we seek to solve the optimization problem:

$$\min_{\mathbf{w}} ||\mathbf{y} - \mathbf{X}\mathbf{w}||^2 \tag{1}$$

  also known as the *minimization of the sum of squares*

Linear Regression error function





Figures taken from [3] and [1]

Minimizing the sum of squares

- Can be done by...

Minimizing the sum of squares

- Can be done by... deriving and setting to zero

$$
\begin{aligned}
\nabla E(\mathbf{w}) &= \nabla(\mathbf{y}-\mathbf{X}\mathbf{w})^T(\mathbf{y}-\mathbf{X}\mathbf{w}) \\
&= \nabla(\mathbf{y}^T\mathbf{y}-\mathbf{y}^T\mathbf{X}\mathbf{w}-\mathbf{w}^T\mathbf{X}^T\mathbf{y}+\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}) \\
&= \nabla(\mathbf{y}^T\mathbf{y}-\mathbf{y}^T\mathbf{X}\mathbf{w}-\mathbf{y}^T\mathbf{X}\mathbf{w}+\mathbf{w}^T\mathbf{X}^T\mathbf{X}\mathbf{w}) \\
&= -2\mathbf{X}^T\mathbf{y}+(\mathbf{X}^T\mathbf{X}+(\mathbf{X}^T\mathbf{X})^T)\mathbf{w} \\
&= -2\mathbf{X}^T\mathbf{y}+2(\mathbf{X}^T\mathbf{X})\mathbf{w} \\
&= -2\mathbf{X}^T(\mathbf{y}-\mathbf{X}\mathbf{w})
\end{aligned}
$$

- Thus:

$$
\begin{aligned}
\nabla E(\mathbf{w}) = 0 \quad &\Rightarrow \quad \mathbf{X}^T\mathbf{X}\mathbf{w} = \mathbf{X}^T\mathbf{y} \\
&\Rightarrow \quad \text{assuming that } \mathbf{X}^T\mathbf{X} \text{ is invertible (non-singular)} \Rightarrow \\
&\Rightarrow \quad \mathbf{w} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}
\end{aligned}
$$

Comments on the solution

- What can we say about the gradient of the loss function ?

$$\nabla E(\mathbf{w}) \quad = \quad -2\mathbf{X}^T(\mathbf{y} - \mathbf{X}\mathbf{w})$$

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  It is a weighted sum of the learning instances where each instance contributes according to the error.
- Even though we have a closed form solution we can also do gradient descent, batch or on-line, in the later case the updates would be:

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$$\mathbf{x}(y - \mathbf{x}^T\mathbf{w})$$

- $\mathbf{X}^\dagger = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is a particular *left inverse* of $\mathbf{X}$ which we call *pseudoinverse*
- Why could it make sense to do gradient descent instead of getting the closed form solution?

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Comments

- What can we say about the $(\mathbf{X}^T\mathbf{X})$ matrix? it is the covariance matrix if the data are centered (i.e. mean removed)
- If the columns of $\mathbf{X}$ are not linearly independent (when does that happen?) then:
  - $\mathbf{X}$ is not of full rank
  - $\mathbf{X}^T\mathbf{X}$ is singular
  - we do not have a unique solution $\mathbf{w}$
- The same rank deficiency problems we have when $d > n$
- In both cases above the solution is either to remove features or to use *model regularization*

Linear regression, a geometric interpretation, I

- Given the training matrix $\mathbf{X} : n \times d$ and the linear model $\mathbf{Xw}$
- The d *columns* ($n \times 1$ dimensional vectors; what the columns correspond to?) of the $\mathbf{X}$ matrix
  - act as the basis of a d-dimensional linear subspace (the column space of $\mathbf{X}$, noted by $C(\mathbf{X})$
  - which is embedeed in the R^n dimensional space
- We have:
  - $\mathbf{y} \in R^n$
  - $\hat{\mathbf{y}} = \mathbf{Xw} \in C(\mathbf{X})$. Why?
- In fact $\hat{\mathbf{y}}$ is the vector in $C(\mathbf{X})$ that has the minimum distance from $\mathbf{y}$, i.e. it is the projection of $\mathbf{y}$ on $C(\mathbf{X})$. Why?

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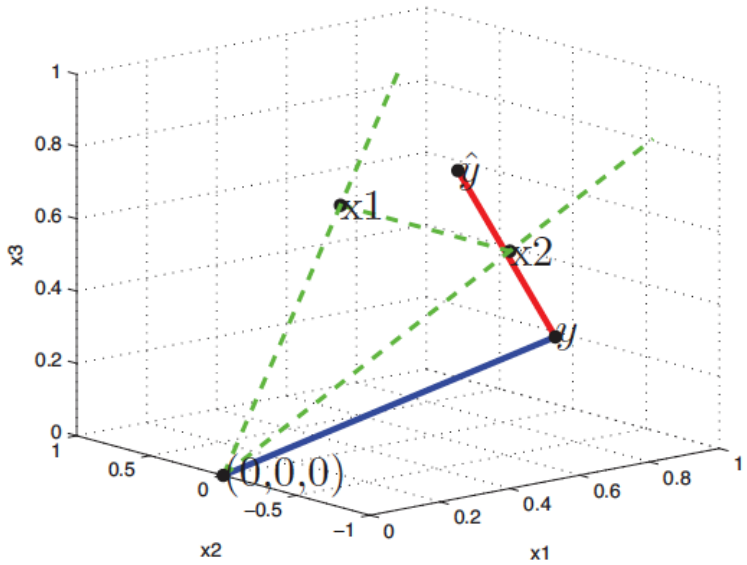
- In fact $\hat{\mathbf{y}}$ is the vector in $C(\mathbf{X})$ that has the minimum distance from $\mathbf{y}$, i.e. it is the projection of $\mathbf{y}$ on $C(\mathbf{X})$. Why? because it is the solution of:

$$\min_{\mathbf{w}} ||\mathbf{y} - \mathbf{Xw}||^2$$

- $\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$: the projection of $\mathbf{y}$ on $C(\mathbf{X})$

Linear regression, a geometric interpretation, II, [3]

- Let $\mathbf{X} = \begin{pmatrix} 1 & 2 \\ 1 & -2 \\ 1 & 2 \end{pmatrix} : n = 3, d = 2, \mathbf{y} = \begin{pmatrix} 8.8957 \\ 0.6130 \\ 1.7761 \end{pmatrix}$



- The two features x_1 and x_2 of $\mathbf{X}$ define the green two-dimensional hyperplane which is the $C(\mathbf{X})$ space
- $\mathbf{y} \in R^3, \hat{\mathbf{y}} \in C(\mathbf{X})$, $\hat{\mathbf{y}} - \mathbf{y}$ the residual vector for which $\min_{\mathbf{w}} ||\hat{\mathbf{y}} - \mathbf{y}||^2$

Linear regression, a parametric model

- We can also derive the linear regression optimization problem under some specific model assumptions
- We assume that the target value y is a linear function of $\mathbf{x}$ plus some noise ϵ, i.e.

$$y = \mathbf{w}^T\mathbf{x} + \epsilon$$

where $\epsilon \sim N(0, \sigma^2)$ or:

$$p(y|\mathbf{x}, \boldsymbol{\theta}) = N(y|\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$$

- The simplest assumption is:

$$\mu(\mathbf{x}) = \mathbf{w}^T\mathbf{x}, \sigma^2(\mathbf{x}) = \sigma^2$$

- As usual we will assume that the training examples are independent and indentically distributed.

Linear regression, a parametric model: the error function, I

- We will use the...

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$$
\begin{aligned}
E(\mathbf{w}) = p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) &= \prod_{i=1}^{i=n} p(y_i|\mathbf{x}_i, \boldsymbol{\theta}) = \prod_{i=1}^{i=n} N(y|\mu(\mathbf{x}), \sigma^2(\mathbf{x})) \\
&= \prod_{i=1}^{i=n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{1}{2\sigma^2}(y_i - \mathbf{w}^T\mathbf{x}_i)^2)
\end{aligned}
$$

- and for convenience the negative log-likelihood:

$$
\begin{aligned}
-\ln(E(\mathbf{w})) &= -\sum_{i=1}^{N} \ln[\frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{1}{2\sigma^2}(y_i - \mathbf{w}^T\mathbf{x}_i))] \\
&= -\sum_{i=1}^{N} \ln(\frac{1}{\sqrt{2\pi\sigma^2}}) + \sum_{i=1}^{N} \frac{1}{2\sigma^2}(y_i - \mathbf{w}^T\mathbf{x}_i)^2
\end{aligned}
$$

Linear regression, a parametric model: the error function, II

- So minimizing the negative likelihood gives:

$$\begin{aligned}
\arg\min_{\mathbf{w}} -\ln(E(\mathbf{w})) &= \arg\min_{\mathbf{w}} -\sum_{i=1}^{N} \ln(\frac{1}{\sqrt{2\pi\sigma^2}}) \\
&+ \sum_{i=1}^{N} \frac{1}{2\sigma^2}(y_i - \mathbf{w}^T\mathbf{x}_i)^2 \\
&= \arg\min_{\mathbf{w}} \sum_{i=1}^{N} (y_i - \mathbf{w}^T\mathbf{x}_i)^2
\end{aligned}$$

- which is a minimization of a sum of squares problem.

Rank deficiencies for $\mathbf{X}$ and regularization

- When we have redundant features the rank of $\mathbf{X}$ is lower than d and $\mathbf{X}^T\mathbf{X}$ is not invertible
- or when $d > n$ then the rank of $\mathbf{X}$ is at most n and as before $\mathbf{X}^T\mathbf{X}$ is not invertible
- In both cases the linear system

$$\mathbf{X}^T\mathbf{X}\mathbf{w} = \mathbf{X}^T\mathbf{y}$$

  is underdetermined (more unknowns than independent equations) and does not have a unique solution.
- For example when we have two perfectly correlated attributes a large positive coefficient in one of them can be compensated by a large negative in the other. The w_i can have high variance, small changes in training set large changes in w_is.

Regularization, ℓ_2 norm, ridge regression, I

- We want to get a unique solution
- One way to do that is to control the size of the w_is by defining the constrained optimization problem:

$$\min_{\mathbf{w}} \quad \sum_{i=1}^{N} (y_i - (\mathbf{w}^T \mathbf{x}_i + w_0))^2 \tag{2}$$

$$\text{subject to} \quad \sum_{j=1}^{d} w_i^2 = \ell_2^2(\mathbf{w}) \leq s$$

  note that now we do not include w_0 in $\mathbf{w}$, the norm is only applied on the coefficients not on the distance of the hyperplane form the origin
- The term $\sum_{j=1}^{d} w_i^2$ is a model complexity term called a regularizer.

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- The optimization problem 2 can be written as

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- The optimization problem 2 can be written as

$$\min_{\mathbf{w}} E_{\ell_2}(\mathbf{w}) \quad = \quad \sum_{i=1}^{N}(y_i - (\mathbf{w}^T\mathbf{x}_i + w_0))^2 + \lambda \sum_{j=1}^{d} w_i^2 \quad (3)$$

  using the Langrange method, $\lambda \geq 0$. Optimization problems 2 and 3 are known as ridge regression.
- They have a closed form solution, found by deriving eq 3 and setting to zero, which is given by:

$$\mathbf{w} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{y} \quad (4)$$

  for ridge regression ([1], section 3.4.1).

Model complexity regularization

- The cost function of Ridge regression:

$$E_{\ell_2}(\mathbf{w}) = \sum_{i=1}^{N} (y_i - (\mathbf{w}^T \mathbf{x}_i + w_0))^2 + \lambda \sum_{j=1}^{d} w_i^2$$

is a trade off between an error component and a model complexity component:

$$\min_{\mathbf{w}} \qquad \text{Error}(\mathbf{w}) + \lambda \text{complexity}(\mathbf{w})$$

such trade-offs are used all the time in learning

Model complexity regularization

- A complex model can fit perfectly any training dataset but will be useless outside it.
- We need to find the right balance between training data fit (error) and model complexity

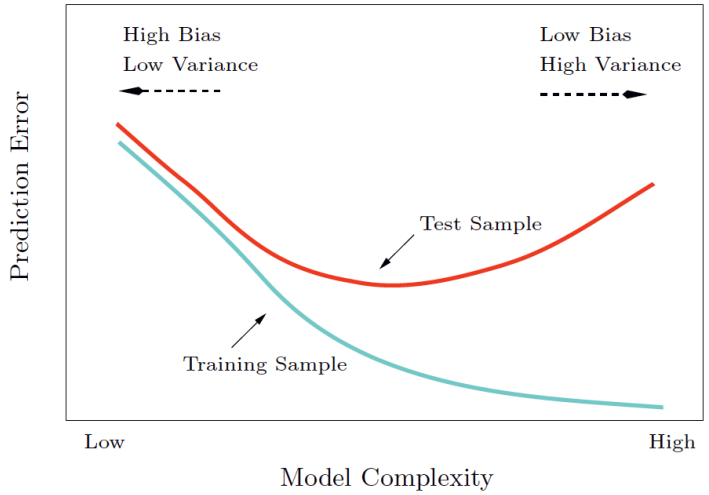

Figure: Training and testing as a function of model complexity, [1]

Ridge regression, comments, I

- The effect of:
  - the constraint $\sum_{j=1}^{d} w_i^2 \leq s$ as $s \to 0$
  - or equivalently of the term $\lambda \sum_{j=1}^{d} w_i^2$ as $\lambda \to \infty$
- is to shrink the coefficients towards zero, i.e. simplify the model.
- How does the Ridge optimization problem, eq 2, look like?

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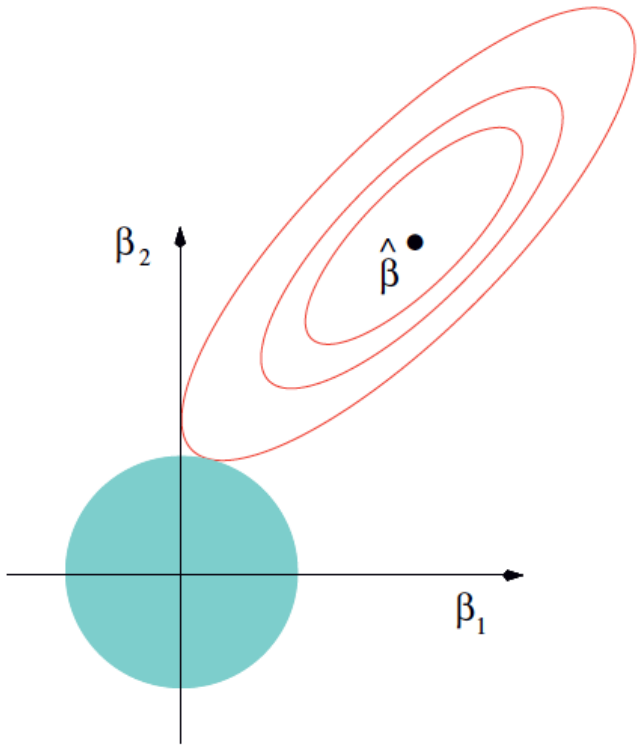
Ridge regression optimization problem visualization: In red the contours of the error function, sum of squares, in turquoise the model norm ball given by s ($\boldsymbol{\beta} := \mathbf{w}$).



Ridge regression, comments, II

- $(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T$ recovers the pseudoinverse of $\mathbf{X}$ as $\lambda \to 0$.
- $(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})$ is always invertible, even if $\mathbf{X}^T\mathbf{X}$ is not, and was the original motivation of ridge regression
- Ridge solutions are not equivariant under scaling of the inputs so we need to standardize the inputs before solving 3
- If we had add w_0, the intercept, in the regularizer would make the procedure depend on the origin chosen for $\mathbf{y}$, i.e. adding a constant to each of the targets y_i would not simply result in a shift of the predictions by the same constant.

Other regularizers

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- A regularization term controls the structure of the model and there are many ways to do that
- Given Ridge regression the most obvious next choise is... $\ell_1(\mathbf{w})$ regularization:

$$\ell_1(\mathbf{w}) = \sum_{i=1}^{d} |w_i|$$

- The optimization problem

$$\begin{aligned} \min_{\mathbf{w}} \quad & \sum_{i=1}^{N} (y_i - (\mathbf{w}^T \mathbf{x}_i + w_0))^2 \\ \text{subject to} \quad & \sum_{j=1}^{d} |w_i| = \ell_1(\mathbf{w}) \leq s \end{aligned} \tag{5}$$

Other regularizers

- or its equivalent form:

$$\min_{\mathbf{w}} E_{\ell_1}(\mathbf{x}) \quad = \quad \sum_{i=1}^{N}(y_i - (\mathbf{w}^T\mathbf{x}_i + w_0))^2 + \lambda \sum_{j=1}^{d} |w_i| \quad (6)$$

  is known as the *lasso* regression.
- A main difference from ridge regression is that the ℓ_1 penalty will force many coefficients w_i to be exactly zero, thus performing feature selection.

Lasso regression, comments

- The effect of:
  - the constraint $\sum_{j=1}^{d} |w_i| \leq s$ as $s \to 0$
  - or equivalently of the term $\lambda \sum_{j=1}^{d} |w_i|$ as $\lambda \to \infty$
- is to shrink more and more coefficients *exactly to* zero, i.e. essentially performing feature selection.
- How does the Lasso optimization problem, eq 2, look like?

Lasso regression, comments

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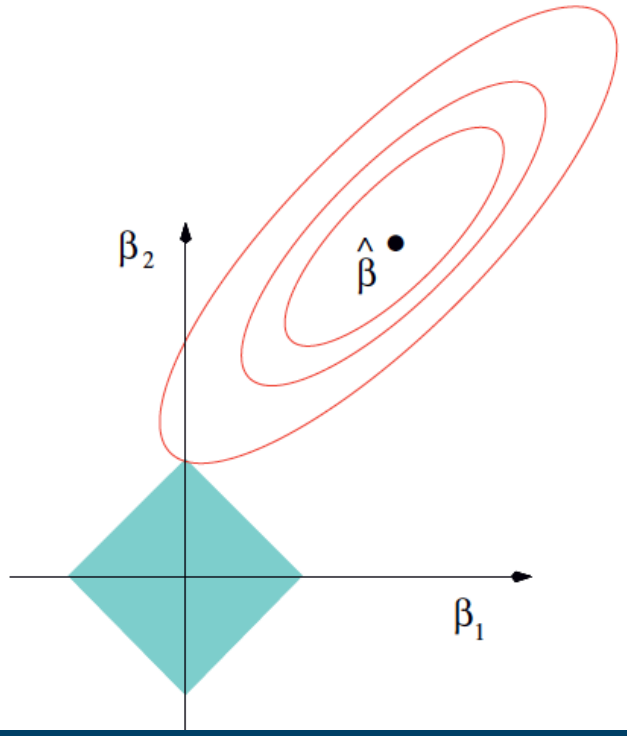
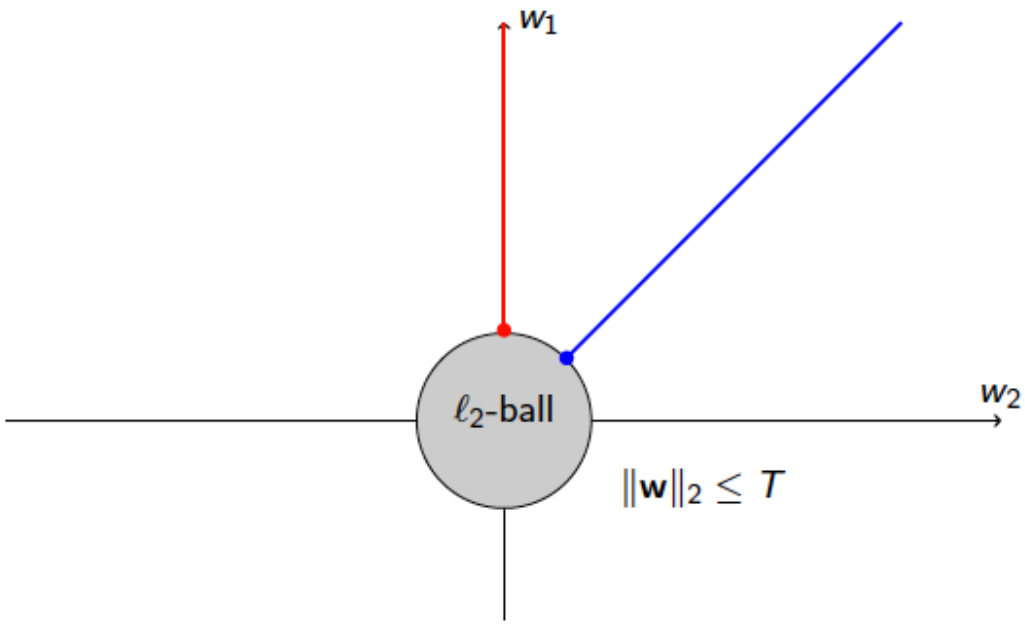
Lasso regression optimization problem visualization: In red the contours of the error function, sum of squares, in turquoise the model norm ball given by s ($\boldsymbol{\beta} := \mathbf{w}$).



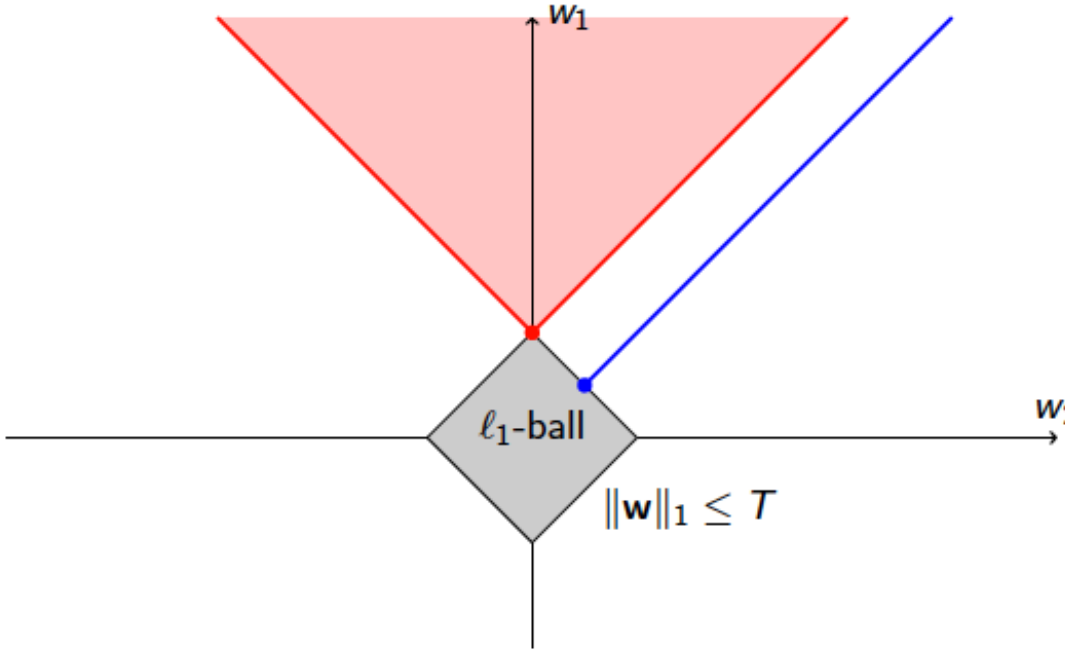
How Ridge and Lasso Actually Behave, I, [2]

Under Ridge the w_2 coefficient will be *exactly* zero only when the optimal solution lies on the w_1 axis



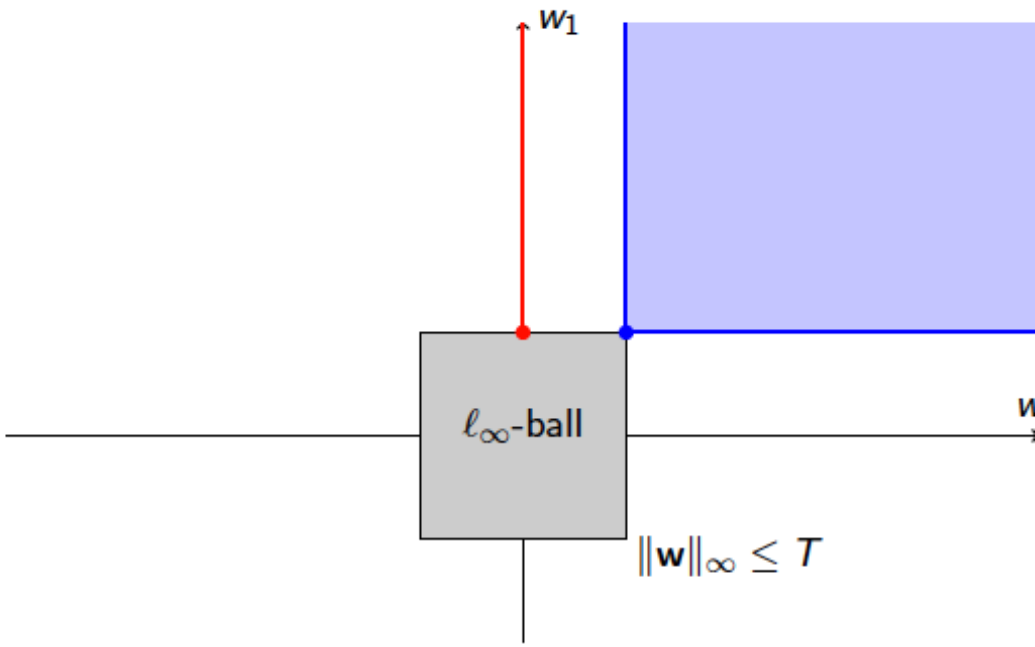
How Ridge and Lasso Actually Behave, II

Under Lasso the w_2 coefficient will be *exactly* zero when the optimal solution lies in the red area



How the ℓ_∞ Behaves

- $\ell_\infty(\mathbf{w}) = \max_{w_i}\{|w_i| | i = 1 \ldots d\}$
- It encourages $|w_1| = |w_2|$



How to solve the lasso optimization problem

- So how do we go about solving:

$$\min_{\mathbf{w}} \quad \sum_{i=1}^{N}(y_i - (\mathbf{w}^T\mathbf{x}_i + w_0))^2 + \lambda \sum_{j=1}^{d} |w_i| \tag{7}$$

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- take the gradient and set to zero does not work... Why? $|x|$ is non-differentiable.
- Here comes in the subgradient

The subgradient

- The subgradient is the extension of gradient to *convex* but *non-differentiable* functions
- Given a *convex* function $f : R^p \rightarrow R$ and a vector $\mathbf{x}_i \in R^p$ a vector $\mathbf{z}$ is called *subgradient* at $\mathbf{x}_i$ if:

$$f(\mathbf{x}_i) + \mathbf{z}^T(\mathbf{x} - \mathbf{x}_i) \leq f(\mathbf{x}), \ \forall \mathbf{x} \in R^p$$

- Note that:

$$f(\mathbf{x}_i) + \mathbf{z}^T(\mathbf{x} - \mathbf{x}_i)$$

is the...

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- Note that:

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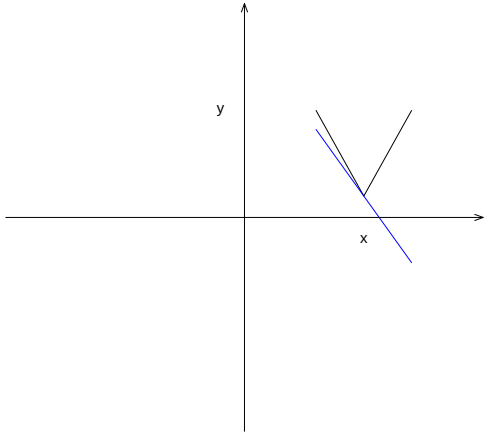
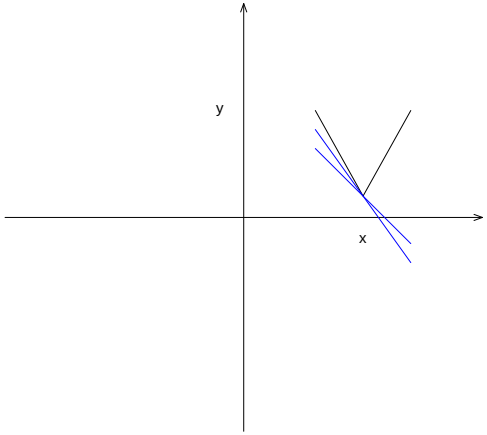
  is the... line that passes from $(\mathbf{x}_i, f(\mathbf{x}_i))$ and has a slope of $\mathbf{z}$
- remember that the tangent at $(\mathbf{x}_i, f(\mathbf{x}_i))$ is the line:

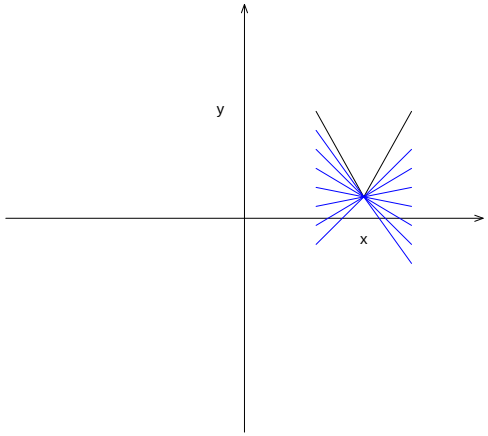
$$f(\mathbf{x}_i) + \nabla f(\mathbf{x}_i)^T(\mathbf{x} - \mathbf{x}_i)$$

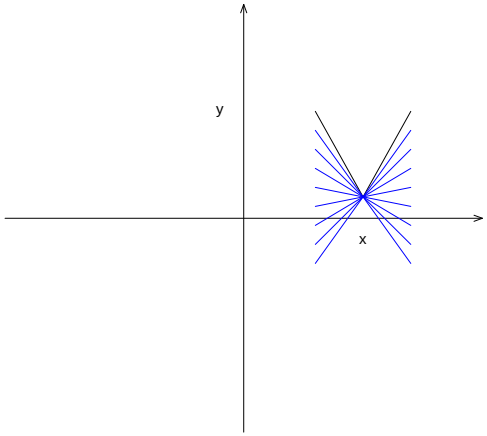
  for which if $f(\mathbf{x})$ is convex it holds that $\leq f(\mathbf{x})$.

Subgradients of $f(x) = \alpha|x - c| + b$ at $x = c$ (non-diff. point)









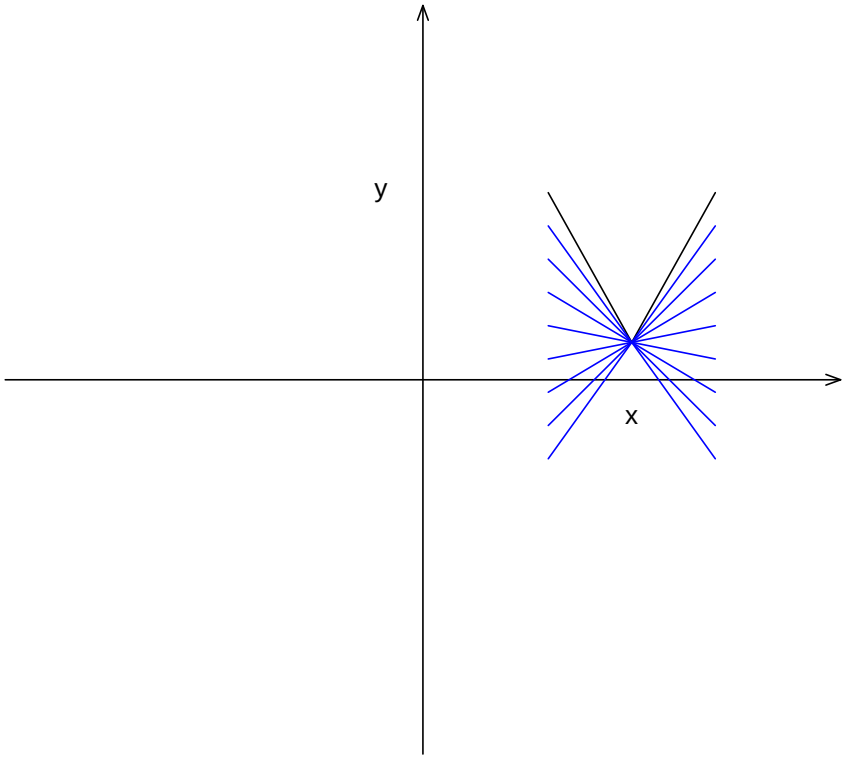
The subdifferential I

- Given a *convex* function $f : R^p \to R$ and a vector $\mathbf{x}_i \in R^p$ the *subdifferential* of f at $\mathbf{x}_i$ is defined as the *set*:

$$\partial f(\mathbf{x}_i) := \{\mathbf{z} \in R^p | f(\mathbf{x}_i) + \mathbf{z}^T(\mathbf{x} - \mathbf{x}_i) \leq f(\mathbf{x}),\ \forall \mathbf{x} \in R^p\}$$

  i.e. it is the set of all subgradient vectors $\mathbf{z}$
- Thus the subdifferential at a point $\mathbf{x}_i$ is a *set*
- so the subdifferential of a convex function at a point $\mathbf{x}_i$ is the set of all slope vectors $\mathbf{z}$ the lines of which pass from $(\mathbf{x}_i, f(\mathbf{x}_i))$ and are bellow the $f(\mathbf{x})$ for every value of $\mathbf{x}$

A subset of the subdifferential of $f(x) = \alpha|x - c| + b$ at $x = c$ (non-diff. point)



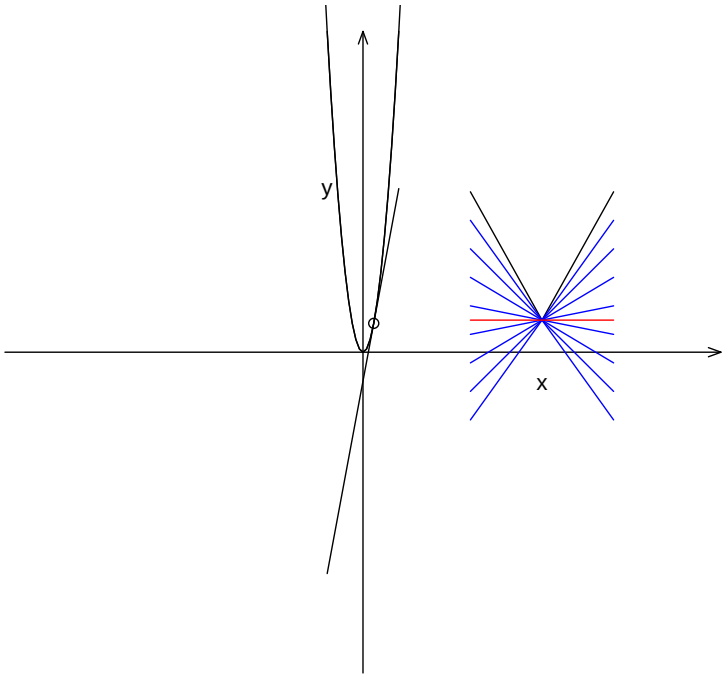
Subgradients at optimality I

- For any *convex* function $f : R^p \to R$, a point $\mathbf{x}_i \in R^p$ is a global minimum if and only if:

$$\mathbf{0} \in \partial f(\mathbf{x}_i)$$

  i.e. the $\mathbf{0}$ vector is in the subdifferential set.
- This is exactly the equivalent of $\nabla f(\mathbf{x}) = \mathbf{0}$ at the minimum of a *convex* function.

Subgradients at optimality II



Plots of $f(x) = x^2$ and its tangent at some point, and $g(x) = a|x - c| + b$ and its subgradients; 0 is in the subdifferential set $[-a, a]$ of $g(x)$ at $x = c$ and thus $g(c)$ is a global minimum.

Subgradient descent for lasso

- So instead of gradiend descent we now do subgradient descent
- And the update rule is:

$$w_{k+1} = w_k - \eta \partial E_{\ell_1}(\mathbf{w})$$

  where η is the learning rate.
- Not necessarily a descent method
  - in a descent method, every step is an improvement
- thus always keep track of the best $\mathbf{w}$ you have seen as we go

How does the Lasso Regression solution relates to the standard linear regression solution

- In the following slides we will see how the Lasso solution relates to the standard linear regression solution.
- We will do so by comparing their optimality conditions, i.e.

$$\nabla E(\mathbf{w}) = 0 \text{ and } \nabla E_{\ell_1}(\mathbf{w}) = 0$$

Lasso optimality conditions I

- The cost function of lasso regression can be written as:

$$E_{\ell_1}(\mathbf{w}) = E(\mathbf{w}) + \lambda \sum_{i=1}^{d} |w_i|$$

where $E(\mathbf{w})$ is the standard regression cost function.

- Remember:

$$\nabla E(\mathbf{w}) = \mathbf{X}^T\mathbf{X}\mathbf{w} - \mathbf{X}^T\mathbf{y}$$

we will denote with:

$$[\nabla E(\mathbf{w})]_k = \frac{\partial E(\mathbf{w})}{\partial w_k}$$

the partial derivative of the standard regression cost function with respect to w_k

Lasso optimality conditions II

- Then the subdifferential of $E_{\ell_1}(\mathbf{w})$ with respect to w_i is given by:

$$\partial_{w_k} E_{\ell_1}(\mathbf{w}) = \frac{\partial E(\mathbf{w})}{\partial w_k} + \lambda \partial_{w_k} |w_k|$$

where:

$$\begin{aligned} \partial_{w_k} |w_k| &= \{-1\} \text{ if } w_k < 0 \\ \partial_{w_k} |w_k| &= [-1, 1] \text{ if } w_k = 0 \\ \partial_{w_k} |w_k| &= \{+1\} \text{ if } w_k > 0 \end{aligned}$$

Lasso optimality conditions III

- Thus:

$$\partial_{w_k} E_{\ell_1}(\mathbf{w}) = \frac{\partial E(\mathbf{w})}{\partial w_k} + \begin{cases} \{-\lambda\} & \text{if } w_k < 0 \\ [-\lambda, \lambda] & \text{if } w_k = 0 \\ \{\lambda\} & \text{if } w_k > 0 \end{cases}$$

and by requiring

$$\mathbf{0} \in \partial_{w_k} E_{\ell_1}(\mathbf{w})$$

we have:

$$\frac{\partial E(\mathbf{w})}{\partial w_k} \in \begin{cases} \{\lambda\} & \text{if } w_k < 0 \\ [-\lambda, \lambda] & \text{if } w_k = 0 \\ \{-\lambda\} & \text{if } w_k > 0 \end{cases} \tag{8}$$

so we need to solve for w_k for the three subcases of eq 8

Lasso optimality conditions IV

- One can show ([3], 13.3)

$$\frac{\partial E(\mathbf{w})}{\partial w_k} = \alpha_k w_k - c_k \tag{9}$$

where:

$$\begin{aligned} \alpha_k &= 2\sum_{i=1}^{n} x_{ik}^2 \\ c_k &= 2\sum_{i=1}^{n} x_{ik}(y_i - \mathbf{w}_{-k}^T \mathbf{x}_{i,-k}) \end{aligned}$$

- c_k is proportional to the correlation of the kth feature to the residual of the model without the kth feature
- *Optimality conditions standard regression*:

$$\frac{\partial E(\mathbf{w})}{\partial w_k} = 0 \Rightarrow w_k = c_k / a_k$$

Lasso optimality conditions V

- Thus from eq 8 and eq 9 we have:

$$
\begin{aligned}
\alpha_k w_k - c_k &\in \begin{cases} \{\lambda\} & \text{if } w_k < 0 \\ [-\lambda, \lambda] & \text{if } w_k = 0 \\ \{-\lambda\} & \text{if } w_k > 0 \end{cases} \\
&\Rightarrow \\
w_k &= \frac{c_k}{\alpha_k} + \frac{\lambda}{\alpha_k} (\text{ if } w_k < 0 \Rightarrow) \text{ when } c_k < -\lambda \\
w_k &= 0 \text{ when } c_k \in [-\lambda, \lambda] \\
w_k &= \frac{c_k}{\alpha_k} - \frac{\lambda}{\alpha_k} (\text{ if } w_k > 0 \Rightarrow) \text{ when } c_k > \lambda
\end{aligned}
$$

Soft thresholding I

- $\mathbf{0} \in \partial_{w_k} E_{\ell_1}(\mathbf{w})$ when:

$$w_k = \frac{c_k}{\alpha_k} + \frac{\lambda}{\alpha_k}(\text{ if } w_k < 0 \Rightarrow) \text{ when } c_k < -\lambda \quad (10)$$

$$w_k = 0 \text{ when } c_k \in [-\lambda, \lambda] \quad (11)$$

$$w_k = \frac{c_k}{\alpha_k} - \frac{\lambda}{\alpha_k}(\text{ if } w_k > 0 \Rightarrow) \text{ when } c_k > \lambda \quad (12)$$

- let:
  - $x_+ = \max(0, x)$ the postitive part of x
  - soft$(\alpha, \delta) = sign(\alpha)(|\alpha| - \delta)_+$: the *soft thresholding* function which shrinks the size of α by δ and if $\alpha < \delta$ sets it to zero.
- then we can rewrite eqs 10,11,12, as:

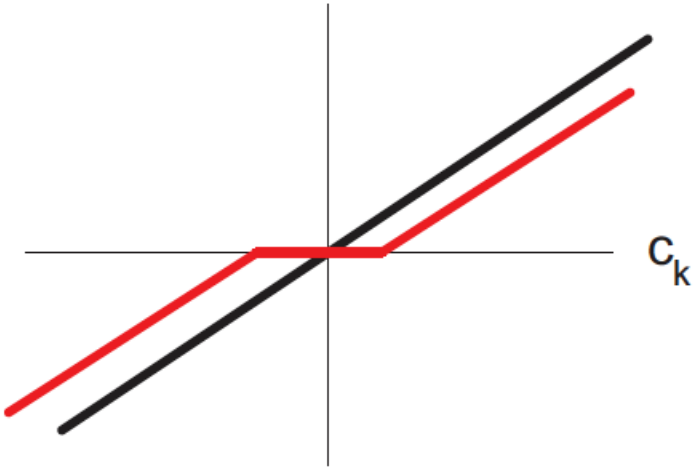
$$w_k = \text{soft}(\frac{c_k}{\alpha_k}, \frac{\lambda}{\alpha_k})$$

Soft thresholding vs hard thresholding I

Plots of w_k vs c_k

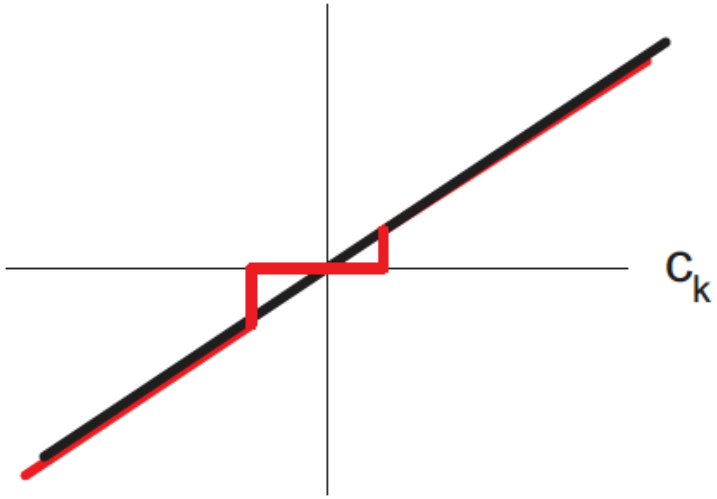
Soft Thresholding

$w_k = \text{soft}(\frac{c_k}{\alpha_k}, \frac{\lambda}{\alpha_k})$



Hard Thresholding

$w_k = 0$ if $c_k \in [-\lambda, \lambda]$



- Black lines: $w_k = c_k/a_k$ solution of standard regression
- Red lines: their sof/hard thresholed variants
- set to zero features with $c_k \in [-\lambda, \lambda]$
- Lasso stands for *least absolute selection and shrinkage operator*

Proximal gradient descent

- Used to optimize objectives of the form:

$$\min_{\mathbf{w} \in R^p} f(\mathbf{w}) + \lambda \Omega(\mathbf{w}) \tag{13}$$

- $f : R^p \rightarrow R$ is a *convex* and *differentiable* function or more formally a generic smooth function with Lipschitz-continuous gradient, i.e.

$$\frac{||f(\mathbf{w}_1) - f(\mathbf{w}_2)||}{||\mathbf{w}_1 - \mathbf{w}_2||} \leq L, \ \forall \mathbf{w}_1, \mathbf{w}_2 \in R^p$$

- $\Omega(\mathbf{w})$ is a *convex* function, often *non-smooth*, typically a sparsity inducing norm, e.g. $\ell_1(\mathbf{w})$
- Quite appropriate method for *large, non-smooth, convex* problems.

Intuition proximal gradient descent

1. First establish a solution vecort $\mathbf{w}_1$ by gradient descent *only* on $f(\mathbf{w})$, i.e. ignore $\Omega(\mathbf{w})$.
2. Get a new $\mathbf{w}_2$ that is close to $\mathbf{w}_1$ and also minimizes $\Omega(\mathbf{w})$
3. Typically $\mathbf{w}_2$ is found through the *proximal operator* of $\Omega(\mathbf{w})$ that often has a *closed form* solution.
4. Iterate 1-3 until convergence

It turns out that the proximal operator of:

$$\Omega(w) = \ell_1(\mathbf{w}) = \sum_i |w_i|$$

denoted by $\mathbf{prox}_{\lambda\ell_1(\mathbf{w})}(\mathbf{x})$ is given by:

$$[\mathbf{prox}_{\lambda\ell_1(\mathbf{w})}(\mathbf{w})]_i = \text{soft}(w_i, \lambda) \quad = \quad \begin{cases} w_i + \lambda & \text{if } w_i < -\lambda \\ 0 & -\lambda < w_i < \lambda \\ w_i - \lambda & \text{if } \lambda < w_i \end{cases}$$

Probabilistic Interpretations

- Both Ridge and Lasso have a probabilistic interpretation
- The solution of the standard linear regression is a maximum likelihood solution
- the solutions of Ridge and Lasso are Maximum a-posteriori estimates with specific priors on the parameters, i.e

$$\arg\max_{\mathbf{w}} p(\mathbf{w}|\mathbf{y}, \mathbf{X}, \lambda) \propto p(\mathbf{y}|\mathbf{X}, \mathbf{w}) P(\mathbf{w}|\lambda)$$

and by taking the negative $\ln$ we have:

$$E_{\ell_2|\ell_1}(\mathbf{w}) = -\ln p(\mathbf{y}|\mathbf{X}, \mathbf{w}) - \ln P(\mathbf{w}|\lambda) \tag{14}$$

$p(\mathbf{y}|\mathbf{X}, \mathbf{w})$ is the likelihood function of standard linear regression

Ridge regression prior and MAP

- The Ridge parameters prior is the Normal prior:

$$p(\mathbf{w}|\lambda) = \prod_{j=1}^{d} N(w_j|0, \lambda^2)$$

- and plugging that prior in eq 14 we get :

$$\begin{aligned} E_{\ell_2}(\mathbf{w}) &= \sum_{i=1}^{n} \frac{1}{2\sigma^2}(y_i - (w_0 + \mathbf{w}^T\mathbf{x}_i))^2 + \frac{1}{\lambda^2}||\mathbf{w}||_2^2 \\ &\propto E(\mathbf{w}) + \lambda'||\mathbf{w}||_2 \end{aligned}$$

$E(\mathbf{w})$ is the sum of squares cost function of linear regression, $\lambda' = \frac{2\sigma^2}{\lambda^2}$.

Lasso regression prior and MAP

- The lasso parameters prior is the Laplace prior:

$$p(\mathbf{w}|\lambda) = \prod_{j=1}^{d} \mathsf{Lap}(w_j|0, 1/\lambda) \propto \prod_{j=1}^{d} \exp(-\lambda|w_j|)$$

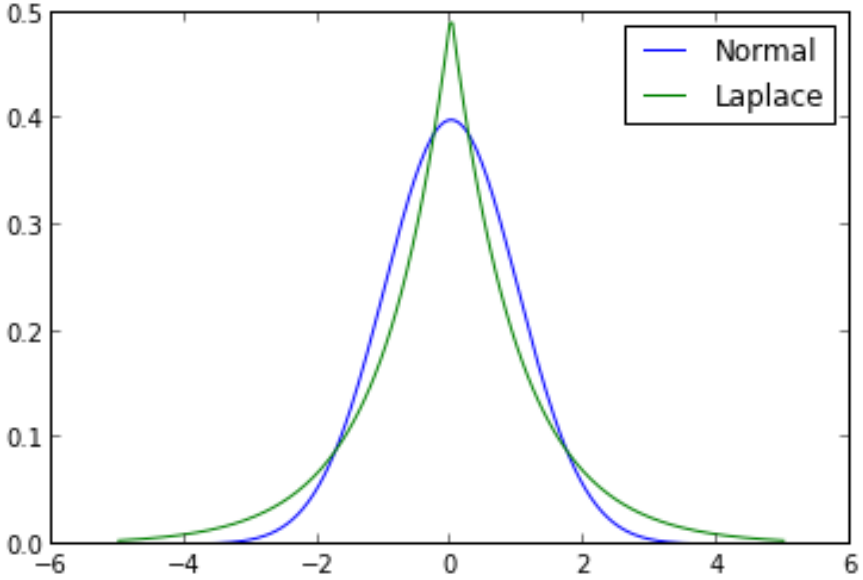
- and plugging that prior in eq 14 we get :

$$\begin{aligned} E_{\ell_1}(\mathbf{w}) &= \sum_{i=1}^{n} \frac{1}{2\sigma^2}(y_i - (w_0 + \mathbf{w}^T\mathbf{x}_i))^2 + \lambda||\mathbf{w}||_1 \\ &\propto E(\mathbf{w}) + \lambda'||\mathbf{w}||_1 \end{aligned}$$

$E(\mathbf{w})$ is the sum of squares cost function of linear regression, $\lambda' = 2\lambda\sigma^2$.

The Laplace distribution

- Lap$(x|\mu, b) := \frac{1}{2b}\exp(-\frac{|x-\mu|}{b})$, μ is a location parameter and $b > 0$ a scale parameter.
- mean $= \mu$, var $= 2b^2$
- Plot of $N(0,1)$, Lap$(0, 1/\sqrt(2)$, $var = 1$ for both.



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